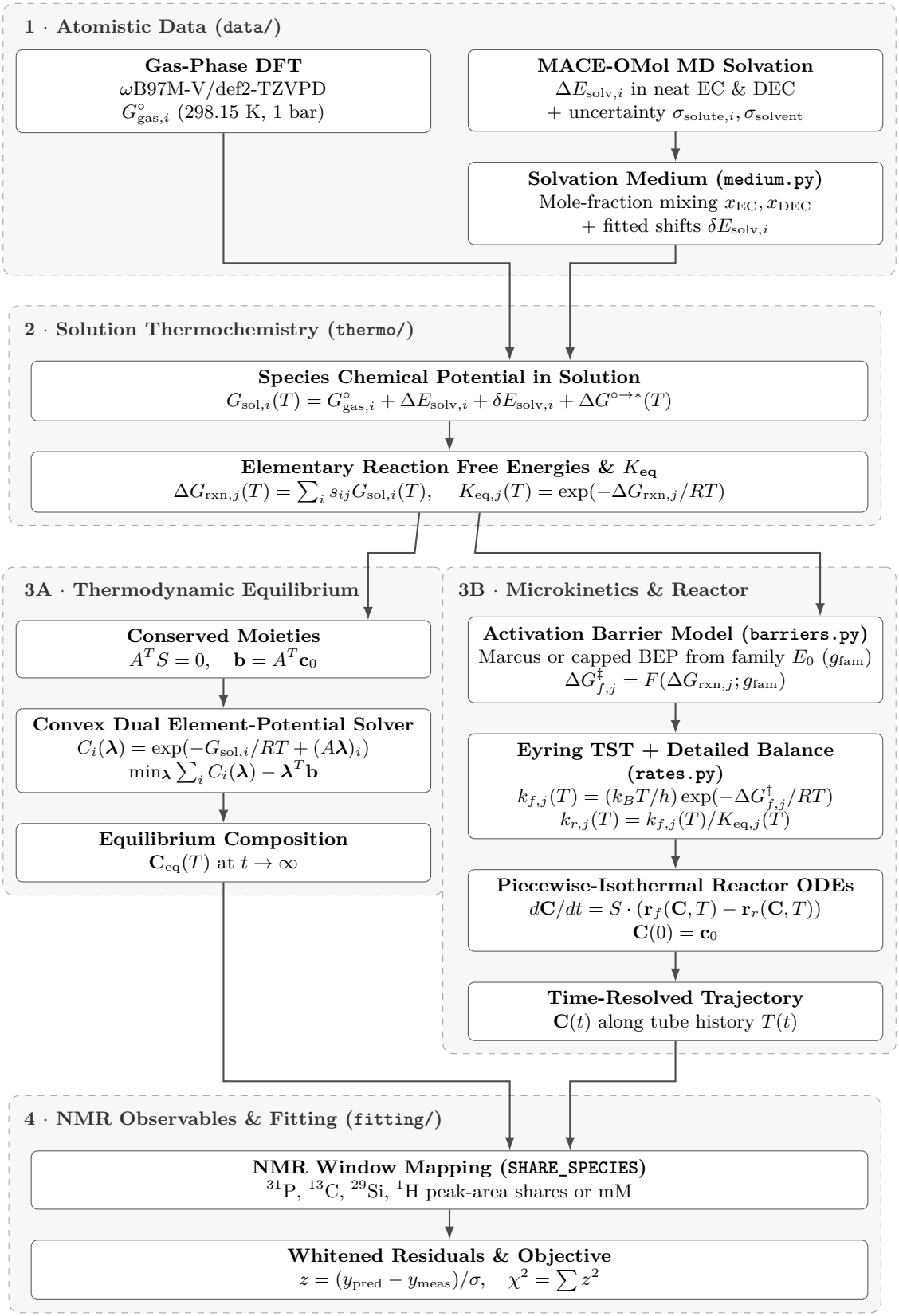
Atom-to-Reactor: Complete Mathematical & Computational Model Reference

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This document provides a complete, equation-by-equation explanation of the **Atom-to-Reactor** multiscale framework implemented in the `kinetics/` package. It traces the full calculation pipeline from gas-phase Density Functional Theory (DFT) and Molecular Dynamics (MD) solvation energies to:

1. **Equilibrium concentrations** $C_i^{\mathrm{eq}}(T)$ via convex thermodynamic Gibbs free-energy minimization (`kinetics/thermo/equilibrium.py`),
2. **Time-resolved concentration trajectories** $C_i(t)$ via transition-state theory, Marcus/BEP activation barrier laws, and stiff mass-action ODE integration (`kinetics/microkinetics/` and `kinetics/reactor/`), and
3. **Experimental NMR observables** (^{31}P, ^{13}C, ^{29}Si, and ^{1}H window shares and millimolar concentrations) used in the thermodynamic and kinetic parameter fits (`kinetics/fitting/`).

1 High-Level Architecture



Codebase Map by Module

Paths are relative to the `kinetics/` package.

Package / Module	Role in the Mathematical Pipeline
`constants.py`	CODATA 2018 fundamental constants (R, k_B, h, N_A), standard states ($P^\circ = 1$ bar, $c^* = 1$ M), and unit conversions ($\mathrm{E_h} \to \mathrm{eV} \to \mathrm{kJ/mol}$).
`data/species.py`	Loads gas-phase DFT energies (`G_gas_eV`, `H_gas_eV`, `E_scf_eV`) and MD solvation energies (`dE_solv_eV`, `dE_solv_sigma_eV`) from the Tank snapshot.
`data/medium.py`	Defines `SolvationMedium`, converting solvent volume fractions into mole fractions x_s and combining neat MD solvation energies.
`data/network.py`	Defines the elementary reaction network (`R1`–`R9` and optional `N10`–`N15`), tracked species, stoichiometric matrix S, and mass-action order matrices.
`thermo/gas.py`	Gas-phase thermochemistry: stored RRHO Gibbs energy at 298.15 K (`wb97mv`) or Grimme quasi-RRHO statistical mechanics (`qRRHO`).
`thermo/standard_state.py`	Standard-state compression shift $\Delta G^{\circ\to *}(T)$ from 1 bar ideal gas to 1 mol/L solution.
`thermo/solution.py`	Solution-phase Gibbs free energy $G_{\mathrm{sol},i}(T)$ (`calculate_solution_gibbs`).
`thermo/reaction.py`	Reaction free energies $\Delta G_{\mathrm{rxn},j}(T)$ and equilibrium constants $K_{\mathrm{eq},j}(T)$ (`calculate_reaction_thermo`).
`thermo/uncertainty.py`	Propagates solute and shared pure-solvent MD uncertainties into reaction covariance Σ and maps reaction shifts to species shifts.
`thermo/equilibrium.py`	Algebraic closed-tube equilibrium solver via convex dual element potentials (`calculate_equilibrium`).
`microkinetics/barriers.py`	Maps $\Delta G_{\mathrm{rxn},j}$ and family intrinsic barrier g (E_0) to forward activation barrier $\Delta G^{\ddagger}_{f,j}$ (`marcus`, `bep_cap`).
`microkinetics/rates.py`	Eyring transition-state theory for $k_{f,j}(T)$, optional Collins–Kimball diffusion ceiling k_D, and detailed balance $k_{r,j}(T) = k_{f,j}(T)/K_{\mathrm{eq},j}(T)$.
`reactor/engine.py`	`MassActionSystem`: net reaction rates $\mathbf{r}(\mathbf{C})$, ODE right-hand side $\frac{d\mathbf{C}}{dt} = S\,\mathbf{r}$, exact analytical Jacobian, and stiff ODE integration (`Radau`/`BDF`).
`reactor/protocol.py`	Piecewise-isothermal multi-stage integration along experimental sample histories (`simulate_history`, `simulate_protocol`).
`fitting/residuals.py`	Maps simulated concentrations to NMR window shares (`SHARE_SPECIES`), applies the noise/baseline/floor error model, and computes whitened residuals.
`fitting/thermo_fit.py`	`ThermoProblem`: equilibrium spectrum selection, replicate compatibility, and fitting species solvation shifts $\delta E_{\mathrm{solv},i}$ to equilibrium NMR data.
`fitting/candidates.py` & `estimation.py`	`FitStructure` and `FitProblem`: kinetic candidate structures (`M0`, `M1`, `M1-split`, `M3-split`), Sobol + TRF optimization, and profile likelihoods.

2 Chemical System & Reaction Network (`kinetics/data/network.py`)

2.1 Tracked Species

When tris(trimethylsilyl) phosphate (**TMSPA**, $OP(OSiMe_3)_3$) scavenges trace water in a carbonate electrolyte (ethylene carbonate **EC** or an **EC:DEC** mixture), it undergoes a cascade of hydrolysis, silanol condensation, trans-silylation, and solvent ring-opening steps. The network tracks $N_{sp} = 11$ reactive species (`NETWORK_SPECIES`):

Group	Species ID	Chemical Formula / Identity	Silyl groups	P atoms
Phosphate ester ladder	`TMSPA`	$OP(OSiMe_3)_3$ — tris(trimethylsilyl) phosphate	3	1
	`BMSPA`	$OP(OH)(OSiMe_3)_2$ — bis(trimethylsilyl) mono-acid phosphate	2	1
	`MMSPA`	$OP(OH)_2(OSiMe_3)$ — mono(trimethylsilyl) di-acid phosphate	1	1
	`H3PO4`	$OP(OH)_3$ — orthophosphoric acid	0	1
Water & free silyl species	`H2O`	H_2O — water	0	0
	`TMSOH`	Me_3SiOH — trimethylsilanol	1	0
	`HMDSO`	$Me_3SiOSiMe_3$ — hexamethyldisiloxane	2	0
Solvent & ring-opening products	`EC`	$(CH_2O)_2CO$ — ethylene carbonate (solvent)	0	0
	`TMSOEG`	$Me_3SiOCH_2CH_2OH$ — 2-(trimethylsiloxy)ethanol	1	0
	`TMSOdiEG`	$Me_3SiOCH_2CH_2OSiMe_3$ — 1,2-bis(trimethylsiloxy)ethane	2	0
	`CO2`	CO_2 — carbon dioxide	0	0

*(Note: Diethyl carbonate **DEC** in 1:1 EC:DEC mixtures is chemically inert in this network and acts as a co-solvent that modifies the solvation medium and dilutes initial molarities.)*

2.2 Elementary Reactions (`R1`–`R9` and `N10`–`N15`)

The default reaction network (`NETWORK` in `kinetics/data/network.py`) comprises 9 elementary reversible steps grouped into 4 reaction families:

ID	Elementary Reaction	Reaction Family (`class`)	Linear Dependency / Cycle Relation
R1	$\mathrm{TMSPA} + \mathrm{H_2O} \rightleftharpoons \mathrm{BMSPA} + \mathrm{TMSOH}$	`hydrolysis`	Independent basis reaction
R2	$\mathrm{BMSPA} + \mathrm{H_2O} \rightleftharpoons \mathrm{MMSPA} + \mathrm{TMSOH}$	`hydrolysis`	Independent basis reaction
R3	$\mathrm{MMSPA} + \mathrm{H_2O} \rightleftharpoons \mathrm{H_3PO_4} + \mathrm{TMSOH}$	`hydrolysis`	Independent basis reaction
R4	$2\,\mathrm{TMSOH} \rightleftharpoons \mathrm{HMDSO} + \mathrm{H_2O}$	`condensation`	Independent basis reaction
R5	$\mathrm{TMSPA} + \mathrm{TMSOH} \rightleftharpoons \mathrm{BMSPA} + \mathrm{HMDSO}$	`transfer`	R5 = R1 + R4
R6	$\mathrm{BMSPA} + \mathrm{TMSOH} \rightleftharpoons \mathrm{MMSPA} + \mathrm{HMDSO}$	`transfer`	R6 = R2 + R4
R7	$\mathrm{MMSPA} + \mathrm{TMSOH} \rightleftharpoons \mathrm{H_3PO_4} + \mathrm{HMDSO}$	`transfer`	R7 = R3 + R4
R8	$\mathrm{EC} + \mathrm{TMSOH} \rightleftharpoons \mathrm{TMSOEG} + \mathrm{CO_2}$	`solvent_attack`	Independent basis reaction
R9	$\mathrm{EC} + \mathrm{TMSOEG} \rightleftharpoons \mathrm{TMSOdiEG} + \mathrm{CO_2}$	`solvent_attack`	Independent basis reaction

In the extended network variants (`'R1-R9+N10-N15'` and `'R1-R9+N10-N15, fixed 1.15 eV'`), 6 additional reactions (`EXTRA_REACTIONS`) between the same 11 species are included:

- **N10** (exchange): $2\,\mathrm{BMSPA} \rightleftharpoons \mathrm{MMSPA} + \mathrm{TMSPA}$ (N10 = R2 − R1)
- **N11** (exchange): $2\,\mathrm{TMSOEG} \rightleftharpoons \mathrm{TMSOH} + \mathrm{TMSOdiEG}$ (N11 = R9 − R8, when coupled with R4)
- **N12** (hydrolysis, follows R2): $\mathrm{BMSPA} + \mathrm{H_2O} \rightleftharpoons \mathrm{H_3PO_4} + \mathrm{HMDSO}$ (N12 = R2 + R7)
- **N13** (exchange): $\mathrm{BMSPA} + \mathrm{H_3PO_4} \rightleftharpoons 2\,\mathrm{MMSPA}$ (N13 = R2 − R3)
- **N14** (exchange): $\mathrm{BMSPA} + \mathrm{MMSPA} \rightleftharpoons \mathrm{H_3PO_4} + \mathrm{TMSPA}$ (N14 = R3 − R1)
- **N15** (hydrolysis, follows R1): $\mathrm{TMSPA} + \mathrm{H_2O} \rightleftharpoons \mathrm{MMSPA} + \mathrm{HMDSO}$ (N15 = R1 + R6)

2.3 Stoichiometric Matrix S and Reaction Order Matrices

For a network of N_{sp} species and N_{rxn} reactions, `build_stoichiometric_matrix` constructs the matrix $S \in \mathbb{R}^{N_{\mathrm{sp}} \times N_{\mathrm{rxn}}}$ with entries:

$$s_{i,j} = \nu_{i,j}^{\mathrm{prod}} - \nu_{i,j}^{\mathrm{react}} \tag{1}$$

where $\nu_{i,j}^{\mathrm{react}} \geq 0$ and $\nu_{i,j}^{\mathrm{prod}} \geq 0$ are the forward and reverse molecularities returned by `build_order_matrices`.

3 From Atomistic Energies (DFT & MD) to Reaction Thermodynamics (`kinetics/thermo/`)

Both the thermodynamic equilibrium solver and the microkinetic ODE solver start from the exact same species chemical potentials in solution $G_{\mathrm{sol},i}(T)$ and elementary reaction free energies $\Delta G_{\mathrm{rxn},j}(T)$.

Step 3.1 · Gas-Phase DFT Gibbs Energy (`kinetics/thermo/gas.py`)

For each species i, the Tank dataset snapshot stores gas-phase DFT calculations at the ω**B97M-V/def2-TZVPD** level of theory ($T_{\mathrm{std}} = 298.15$ K, $P^\circ = 1$ bar $= 10^5$ Pa). Two thermochemical modes (`thermo_mode`) are supported in `calculate_gas_thermo`:

1. **`'wb97mv'` (Default across fits):** Uses the stored rigid-rotor harmonic-oscillator (RRHO) Gibbs free energy converted from Hartree to eV and kJ/mol (1 $\mathrm{E_h} = 27.211386$ eV, 1 eV $= 96.485332$ kJ/mol):
$$G^\circ_{\mathrm{gas},i} = E^{\mathrm{Eh}}_{\mathrm{gas},i} \times 27.211386 \text{ eV/E}_\mathrm{h} \tag{2}$$
2. **`'qRRHO'` (Grimme quasi-RRHO):** When vibrational frequencies $\{\tilde\nu_m\}$ and principal moments of inertia are present, computes translational (Sackur–Tetrode at $P^\circ = 1$ bar), rigid-rotor rotational, and damped quasi-RRHO vibrational enthalpies and entropies at temperature T with damping frequency $\tilde\nu_0 = 100$ cm^{-1}:
$$w(\tilde\nu_m) = \frac{1}{1 + (\tilde\nu_0/\tilde\nu_m)^4}, \qquad S_{\mathrm{vib}} = \sum_m \left[w(\tilde\nu_m) S_{\mathrm{HO}}(\tilde\nu_m, T) + (1 - w(\tilde\nu_m)) S_{\text{free-rot}}(\tilde\nu_m, T) \right] \tag{3}$$

Step 3.2 · Solvation Energy in Neat or Mixed Solvents (`kinetics/data/medium.py`)

Molecular dynamics (MD) simulations using the MACE-OMol machine-learned interatomic potential provide the solvation energy $\Delta E^{(s)}_{\mathrm{solv},i}$ of each solute i in neat solvent $s \in \{\mathrm{EC}, \mathrm{DEC}\}$.

For a solvent mixture specified by volume fractions $\{f_s\}$ (such as **EC:DEC 1:1 by volume** in the lab NMR tubes), `SolvationMedium.mole_fractions` converts volume fractions into mole fractions x_s using the neat liquid densities ($\rho_{\mathrm{EC}} = 1.321$ g/mL, $\rho_{\mathrm{DEC}} = 0.975$ g/mL) and molar masses M_s:

$$n_s = \frac{f_s\, \rho_s}{M_s}, \qquad x_s = \frac{n_s}{\sum_{s'} n_{s'}} \qquad (\text{for EC:DEC 1:1 v/v: } x_{\mathrm{EC}} \approx 0.645,\ x_{\mathrm{DEC}} \approx 0.355) \tag{4}$$

Under the linear ideal-mixing rule (`_cached_medium_database`), the raw solvation energy of species i in the mixture is the mole-fraction-weighted sum:

$$\Delta E^{\mathrm{raw}}_{\mathrm{solv},i} = \sum_s x_s\, \Delta E^{(s)}_{\mathrm{solv},i} \tag{5}$$

When species free-energy shifts $\delta E_{\mathrm{solv},i}$ are introduced (either during a thermodynamic fit in `thermo_fit.py` or via freed reaction energies in `candidates.py`), the effective solvation energy becomes:

$$\Delta E_{\mathrm{solv},i} = \Delta E^{\mathrm{raw}}_{\mathrm{solv},i} + \delta E_{\mathrm{solv},i} \tag{6}$$

Step 3.3 · Standard-State Compression Shift (`kinetics/thermo/standard_state.py`)

Gas-phase DFT quantities are referenced to an ideal gas at $P^\circ = 1$ bar with molar concentration $c^\circ_{\text{gas}}(T) = \frac{P^\circ}{RT}$, whereas solution-phase thermodynamics and mass-action kinetics use a $c^* = 1$ mol/L $= 1000$ mol/m^3 reference state. Compressing 1 mole of ideal gas into 1 L of solution adds the entropic concentration shift (`calculate_standard_state_shift`):

$$\Delta G^{\circ\to *}(T) = RT\ln\!\left(\frac{c^*}{P^\circ/(RT)}\right) = RT\ln\!\left(\frac{RTc^*}{P^\circ}\right) \qquad (+7.96\ \text{kJ/mol} = +0.0825\ \text{eV at } 298.15\ \text{K}) \tag{7}$$

(In `'wb97mv'` mode, $\Delta G^{\circ\to *}(T)$ is omitted from individual species energies because every reaction in the default network `R1`–`R9` and `N10`–`N15` is a $2 \to 2$ bimolecular step with $\Delta\nu_j = \sum_i s_{i,j} = 0$, so $\Delta G^{\circ\to *}(T)$ cancels out identically in every $\Delta G_{rxn,j}$.)

Step 3.4 · Solution-Phase Chemical Potential $G_{\mathbf{sol},i}(T)$ (`kinetics/thermo/solution.py`)

Combining the gas-phase Gibbs energy, the solvation energy (plus any fitted shift $\delta E_{\text{solv},i}$), and the standard-state shift yields the standard-state Gibbs energy (standard chemical potential $\mu_i^\circ(T)$) of species i in solution (`calculate_solution_gibbs`):

$$\mu_i^\circ(T) \equiv G_{\text{sol},i}(T) = G^\circ_{\text{gas},i}(T) + \Delta E^{\text{raw}}_{\text{solv},i} + \delta E_{\text{solv},i} + \Delta G^{\circ\to *}(T) \tag{8}$$

Step 3.5 · Elementary Reaction Free Energy $\Delta G_{\mathbf{rxn},j}(T)$ and Equilibrium Constant $K_{\mathbf{eq},j}(T)$ (`kinetics/thermo/reaction.py`)

For each elementary reaction j, `calculate_reaction_thermo` evaluates the standard reaction Gibbs free energy by multiplying $G_{\text{sol},i}(T)$ by the stoichiometric coefficients $s_{i,j} = \nu^{\text{prod}}_{i,j} - \nu^{\text{react}}_{i,j}$:

$$\Delta G_{\text{rxn},j}(T) = \sum_{i=1}^{N_{\text{sp}}} s_{i,j}\, G_{\text{sol},i}(T) = \left(S^T \mathbf{G}_{\text{sol}}(T)\right)_j \tag{9}$$

From $\Delta G_{\text{rxn},j}(T)$, the dimensionless thermodynamic equilibrium constant at temperature T is:

$$K_{\text{eq},j}(T) = \exp\!\left(-\frac{\Delta G_{\text{rxn},j}(T)}{RT}\right) \tag{10}$$

Automatic Wegscheider Cycle Closure: Because $\Delta G_{\text{rxn},j}(T)$ is computed from **species** energies $\mathbf{G}_{\text{sol}}(T)$ via $\Delta\mathbf{G}_{\text{rxn}} = S^T\mathbf{G}_{\text{sol}}$, any closed thermodynamic cycle of reactions $\sum_j c_j s_{i,j} = 0$ automatically satisfies:

$$\sum_j c_j \Delta G_{\text{rxn},j}(T) = 0 \quad\Longleftrightarrow\quad \prod_j \left(K_{\text{eq},j}(T)\right)^{c_j} = 1 \tag{11}$$

In particular, since R5 = R1 + R4, R6 = R2 + R4, and R7 = R3 + R4:

$$\Delta G_{\text{rxn},5} = \Delta G_{\text{rxn},1} + \Delta G_{\text{rxn},4}, \qquad K_{\text{eq},5}(T) = K_{\text{eq},1}(T)\cdot K_{\text{eq},4}(T) \tag{12}$$

and similarly for R6, R7, and all extra steps N10–N15.

Step 3.6 · Solvation Uncertainty Propagation & Reaction Covariance (`kinetics/thermo/uncertainty.py`)

Each MD solvation calculation estimates $\Delta E^{(s)}_{\mathrm{solv},i} = \langle E^{(s)}_{\mathrm{solution},i}\rangle - \langle E_{\mathrm{gas},i}\rangle - \langle E^{(s)}_{\mathrm{solvent}}\rangle$ from three MD trajectories with standard errors of the mean $\sigma_{\mathrm{gas},i}$, $\sigma^{(s)}_{\mathrm{solution},i}$, and $\sigma^{(s)}_{\mathrm{solvent}}$.

1. **Per-species independent vs. shared uncertainty (`calculate_solvation_sigma_components`):** The gas and solution-box trajectories are independent for each solute i, whereas the pure-solvent reference trajectory $\langle E^{(s)}_{\mathrm{solvent}}\rangle$ is shared across all solutes:

$$\sigma^{(s)}_{\mathrm{solute},i} = \sqrt{\left(\sigma_{\mathrm{gas},i}\right)^2 + \left(\sigma^{(s)}_{\mathrm{solution},i}\right)^2}, \qquad \sigma^{(s)}_{\mathrm{tot},i} = \sqrt{\left(\sigma^{(s)}_{\mathrm{solute},i}\right)^2 + \left(\sigma^{(s)}_{\mathrm{solvent}}\right)^2} \tag{13}$$

   In a solvent mixture with mole fractions x_s, these combine as $\sigma_{\mathrm{solute},i} = \sqrt{\sum_s (x_s \sigma^{(s)}_{\mathrm{solute},i})^2}$ and $\sigma_{\mathrm{solvent}} = \sqrt{\sum_s (x_s \sigma^{(s)}_{\mathrm{solvent}})^2}$.

2. **Reaction solvation covariance matrix (`calculate_solvation_covariance`):** Two reactions j and k share species (for example, `BMSPA` and `TMSOH` appear in both R1 and R2), making their solvation errors correlated:

$$\Sigma_{j,k} = \mathrm{Cov}(\Delta\Delta E_{\mathrm{solv},j},\, \Delta\Delta E_{\mathrm{solv},k}) = \sum_{i=1}^{N_{\mathrm{sp}}} s_{i,j}\, s_{i,k}\, \sigma^2_{\mathrm{solute},i} + \left(\sum_i s_{i,j}\right)\left(\sum_i s_{i,k}\right) \sigma^2_{\mathrm{solvent}} \tag{14}$$

   For all $2 \to 2$ reactions ($\sum_i s_{i,j} = 0$), the shared pure-solvent error cancels out completely!

4 Thermodynamics: From $\Delta G_{\mathbf{rxn},j}$ to Equilibrium Concentrations $C_i^{\mathbf{eq}}(T)$ (`kinetics/thermo/equilibrium.py`)

IMPORTANT — What Thermodynamics Computes

The thermodynamic equilibrium model has **no time variable** t and **no activation barriers** E_0. Given the initial concentrations $\mathbf{c}_0$ mixed in a closed NMR tube and a set of reactions that have reached equilibrium at temperature T, it finds the unique composition vector $\mathbf{C}^{\text{eq}}(T)$ at $t \to \infty$ that minimizes the solution Gibbs free energy while conserving all atomic/group moieties.

Step 4.1 · Counting Unknowns vs. Equations (Worked Example: `2 % H2O` Tube)

Suppose we mix the `2 % H2O` tube at $t = 0$ with initial concentrations $\mathbf{c}_0$:

- $C_{\text{TMSPA},0} \approx 0.146$ M (or the tube's exact recipe molarity),
- $C_{\text{H}_2\text{O},0} \approx 1.05$ M,
- $C_{\text{EC},0} \approx 7.1$ M (and $C_{\text{DEC},0} \approx 4.1$ M in EC:DEC 1:1),
- All reaction products start at 0 M.

Case A: Phosphate + Silanol Equilibrium (`R1`–`R4`, and therefore `R5`–`R7` and `N10`, `N12`–`N15`)

In the room-temperature thermodynamic fit (`kinetics/fitting/thermo_fit.py`, where EC ring-opening `R8`–`R9` does not run at room temperature), the active species set spanned by `R1`–`R7` has $N_{\mathbf{sp}} = 7$ **reactive species**:

$$\text{Species} = \{\text{TMSPA},\ \text{BMSPA},\ \text{MMSPA},\ \text{H}_3\text{PO}_4,\ \text{H}_2\text{O},\ \text{TMSOH},\ \text{HMDSO}\} \tag{15}$$

To determine these **7 unknown equilibrium concentrations**, we need **7 independent algebraic equations**:

1. **4 Independent Mass-Action Equilibrium Equations (`R1`–`R4`):**

$$\text{R1:}\quad \frac{C_{\text{BMSPA}} \cdot C_{\text{TMSOH}}}{C_{\text{TMSPA}} \cdot C_{\text{H}_2\text{O}}} = K_{\text{eq},1}(T) = \exp\left(-\frac{\Delta G_{\text{rxn},1}}{RT}\right) \tag{16}$$

$$\text{R2:}\quad \frac{C_{\text{MMSPA}} \cdot C_{\text{TMSOH}}}{C_{\text{BMSPA}} \cdot C_{\text{H}_2\text{O}}} = K_{\text{eq},2}(T) = \exp\left(-\frac{\Delta G_{\text{rxn},2}}{RT}\right) \tag{17}$$

$$\text{R3:}\quad \frac{C_{\text{H}_3\text{PO}_4} \cdot C_{\text{TMSOH}}}{C_{\text{MMSPA}} \cdot C_{\text{H}_2\text{O}}} = K_{\text{eq},3}(T) = \exp\left(-\frac{\Delta G_{\text{rxn},3}}{RT}\right) \tag{18}$$

$$\text{R4:}\quad \frac{C_{\text{HMDSO}} \cdot C_{\text{H}_2\text{O}}}{(C_{\text{TMSOH}})^2} = K_{\text{eq},4}(T) = \exp\left(-\frac{\Delta G_{\text{rxn},4}}{RT}\right) \tag{19}$$

(Why aren't R5, R6, R7 extra equations? Because R5 = R1 + R4, multiplying the equilibrium expressions for R1 and R4 yields $\frac{C_{BMSPA} \cdot C_{HMDSO}}{C_{TMSPA} \cdot C_{TMSOH}} = K_{eq,1} K_{eq,4} = K_{eq,5}$, which is already satisfied!)

2. **3 Independent Atom/Group Conservation Equations ($\mathbf{rank}(S) = 4 \implies \dim(\mathbf{null}(S^T)) = 7 - 4 = 3$):** Because the tube is closed, the total concentrations of phosphorus cores ($-PO_4$), trimethylsilyl groups ($-SiMe_3$), and water/hydroxyl oxygen equivalents are invariant:

   (1) Total phosphorus (b_P) :
   $$C_{\mathrm{TMSPA}} + C_{\mathrm{BMSPA}} + C_{\mathrm{MMSPA}} + C_{\mathrm{H_3PO_4}} = C_{\mathrm{TMSPA},0} \tag{20}$$

   (2) Total silyl groups (b_{Si}) :
   $$3\,C_{\mathrm{TMSPA}} + 2\,C_{\mathrm{BMSPA}} + C_{\mathrm{MMSPA}} + C_{\mathrm{TMSOH}} + 2\,C_{\mathrm{HMDSO}} = 3\,C_{\mathrm{TMSPA},0} \tag{21}$$

   (3) Total water/O equivalents (b_O) :
   $$C_{\mathrm{BMSPA}} + 2\,C_{\mathrm{MMSPA}} + 3\,C_{\mathrm{H_3PO_4}} + C_{\mathrm{H_2O}} + C_{\mathrm{TMSOH}} + C_{\mathrm{HMDSO}} = C_{\mathrm{H_2O},0} \tag{22}$$

   (Note: Any invertible linear combination of these 3 moiety balances spans the same 3D left nullspace $A = null(S^T)$.)

Total: 4 equilibrium laws + 3 conservation laws = **7** equations for **7** unknowns.

Case B: Full 11-Species Network Including Solvent Attack (`R1`–`R9`)

If solvent ring-opening reactions `R8` and `R9` also equilibrate, 4 additional species (`EC`, `TMSOEG`, `TMSOdiEG`, `CO2`) enter, giving $N_{\mathbf{sp}} = 11$ **unknowns**:

- **6 independent equilibrium laws:** $K_{\mathrm{eq},1}, K_{\mathrm{eq},2}, K_{\mathrm{eq},3}, K_{\mathrm{eq},4}, K_{\mathrm{eq},8}, K_{\mathrm{eq},9}$,
- **5 conservation laws ($\dim(\mathbf{null}(S^T)) = 11-6 = 5$):** total phosphorus b_P, total silyl groups b_{Si} (now including $+1\,C_{\mathrm{TMSOEG}} + 2\,C_{\mathrm{TMSOdiEG}}$), total oxygen equivalents b_O, total ethylene glycol rings $C_{\mathrm{EC}} + C_{\mathrm{TMSOEG}} + C_{\mathrm{TMSOdiEG}} = C_{\mathrm{EC},0}$, and decarboxylation stoichiometry $C_{\mathrm{TMSOEG}} + C_{\mathrm{TMSOdiEG}} - C_{\mathrm{CO_2}} = 0$.
- **Total:** $6 + 5 = \mathbf{11}$ equations for **11** unknowns.

Step 4.2 · Convex Dual Element-Potential Solver (`calculate_equilibrium`)

Solving the polynomial mass-action equations directly in concentration space is numerically fragile when some equilibrium constants are very large (e.g., $K_{\mathrm{eq},1} \gg 1$, where $C^{\mathrm{eq}}_{\mathrm{TMSPA}} \sim 10^{-12}$ M in wet tubes). Instead, `calculate_equilibrium` formulates the problem as the **constrained minimization of the ideal-solution Gibbs free energy** in dimensionless units $\tilde{\mu}_i = G_{\mathrm{sol},i}(T)/(RT)$:

$$\min_{\mathbf{C}>\mathbf{0}} \sum_{i=1}^{N_{\mathrm{sp}}} C_i\big(\tilde{\mu}_i + \ln C_i - 1\big) \qquad \text{subject to} \qquad A^T\mathbf{C} = A^T\mathbf{c}_0 \equiv \mathbf{b} \tag{23}$$

where the columns of $A = \mathrm{null}(S^T) \in \mathbb{R}^{N_{\mathrm{sp}} \times N_{\mathrm{cons}}}$ form an orthonormal basis for the left nullspace of the stoichiometric matrix ($S^T A = \mathbf{0}$, $A^T A = I$).

1. **Lagrangian & Dual Parameterization via Element Potentials $\boldsymbol{\lambda}$:** Introducing Lagrange multipliers (element potentials) $\boldsymbol{\lambda} \in \mathbb{R}^{N_{\mathrm{cons}}}$ for the conservation constraints $A^T\mathbf{C} = \mathbf{b}$:
   $$\mathcal{L}(\mathbf{C}, \boldsymbol{\lambda}) = \sum_{i=1}^{N_{\mathrm{sp}}} C_i\big(\tilde{\mu}_i + \ln C_i - 1\big) - \boldsymbol{\lambda}^T\big(A^T\mathbf{C} - \mathbf{b}\big) \tag{24}$$
   Setting $\frac{\partial \mathcal{L}}{\partial C_i} = \tilde{\mu}_i + \ln C_i - (A\boldsymbol{\lambda})_i = 0$ yields the exact exponential representation of every species concentration:
   $$C_i(\boldsymbol{\lambda}) = \exp\big(-\tilde{\mu}_i + (A\boldsymbol{\lambda})_i\big) > 0 \tag{25}$$

2. **Why $C_i(\boldsymbol{\lambda})$ Satisfies Every Equilibrium Constant Automatically:** Taking the reaction stoichiometric combination $\sum_i s_{i,j} \ln C_i(\boldsymbol{\lambda})$ for any reaction j:

$$\sum_{i=1}^{N_{\text{sp}}} s_{i,j} \ln C_i(\boldsymbol{\lambda}) = -\sum_{i=1}^{N_{\text{sp}}} s_{i,j}\tilde{\mu}_i + \left(S^T A\boldsymbol{\lambda}\right)_j = -\frac{\Delta G_{\text{rxn},j}(T)}{RT} + 0 = \ln K_{\text{eq},j}(T) \tag{26}$$

Because $S^T A = \mathbf{0}$, **all mass-action equilibrium equations hold identically for every $\boldsymbol{\lambda} \in \mathbb{R}^{N_{\text{cons}}}$!**

3. **Unconstrained Strictly Convex Dual Optimization:** Substituting $C_i(\boldsymbol{\lambda})$ into the dual objective gives an unconstrained, smooth, strictly convex minimization over only N_{cons} variables (3 variables for `R1`–`R4`, or 5 for `R1`–`R9`):

$$\min_{\boldsymbol{\lambda}\in\mathbb{R}^{N_{\text{cons}}}} \Phi(\boldsymbol{\lambda}) = \sum_{i=1}^{N_{\text{sp}}} \exp\big(-\tilde{\mu}_i + (A\boldsymbol{\lambda})_i\big) - \boldsymbol{\lambda}^T\mathbf{b} \tag{27}$$

with analytical gradient (the conservation residual) and symmetric positive-definite Hessian:

$$\nabla\Phi(\boldsymbol{\lambda}) = A^T\mathbf{C}(\boldsymbol{\lambda}) - \mathbf{b}, \qquad \nabla^2\Phi(\boldsymbol{\lambda}) = A^T \operatorname{diag}(\mathbf{C}(\boldsymbol{\lambda}))\, A \succ 0 \tag{28}$$

`calculate_equilibrium` solves $\nabla\Phi(\boldsymbol{\lambda}^*) = \mathbf{0}$ using damped Newton steps with Armijo backtracking line search until $\|\nabla\Phi\|_\infty \le 10^{-8}\|\mathbf{b}\|_\infty$.

5 Microkinetics & Reactor ODEs: From $\Delta G_{\mathbf{rxn},j}$ and E_0 to Time Trajectories $C_i(t)$ (`kinetics/microkinetics/` & `kinetics/reactor/`)

IMPORTANT — What Kinetics Adds Beyond Thermodynamics

Thermodynamics tells us **where** the system ends up at $t \to \infty$ using only energy differences $\Delta G_{\text{rxn},j}$. Microkinetics introduces **activation barriers** $\Delta G^{\ddagger}_{f,j}$ (parameterized by an intrinsic barrier $g_{\text{fam}} \equiv E_{0,\text{fam}}$ per reaction family) to compute **how fast** each elementary reaction runs at temperature $T(t)$, integrating $\frac{d\mathbf{C}}{dt}$ from $t = 0$ across the experimental temperature history of each NMR tube.

Step 5.1 · Activation Barrier Laws $\Delta G^{\ddagger}_{f,j}$ (`kinetics/microkinetics/barriers.py`)

Let $x_j = \Delta G_{\text{rxn},j}(T)$ (in eV) be the reaction free energy of step j, and let $g_j = E_{0,\text{fam}(j)} > 0$ be the intrinsic barrier of its reaction family (the barrier when $\Delta G_{\text{rxn}} = 0$). If a family has a non-zero activation entropy $\Delta S^{\ddagger}$, the intrinsic barrier at temperature T is:

$$g_j(T) = g_j(T_{\text{ref}}) - (T - T_{\text{ref}})\,\Delta S^{\ddagger}_{\text{fam}(j)} \tag{29}$$

Two barrier shapes are registered in `BARRIER_MODELS`:

1. **Marcus Group-Transfer Law (`'marcus'`, Default in `level1`, `M0-Marcus`, `M1`, `M1-split`, `M3-split`):** With reorganization energy $\lambda_j = 4g_j$, the quadratic Marcus barrier and its effective Brønsted/Leffler slope $\alpha_{\text{eff}} = \frac{\partial \Delta G^{\ddagger}_f}{\partial x}$ are (`marcus` and `calculate_barrier`):

$$\begin{aligned}\Delta G^{\ddagger}_{f,j}(x_j, g_j) &= \max\left(0,\ x_j,\ g_j\left(1 + \frac{x_j}{4g_j}\right)^2\right) \\ &= \begin{cases} 0 & \text{if } x_j \le -4g_j \text{ (barrierless exergonic)} \\ g_j + \dfrac{x_j}{2} + \dfrac{x_j^2}{16 g_j} & \text{if } |x_j| < 4g_j \\ x_j & \text{if } x_j \ge +4g_j \text{ (endergonic floor)} \end{cases}\end{aligned} \tag{30}$$

   (Optional work terms w_R, w_P shift the parabola via $\Delta G^{\ddagger}_{f,j} = \max(0,\ x_j,\ w_R + F_{Marcus}(x_j - w_R + w_P; g_j))$. Notice that Marcus satisfies reversal invariance $F(-x; g) = F(x; g) - x$, meaning forward and reverse barriers always differ by $x_j = \Delta G_{rxn,j}$ regardless of the direction in which the reaction is written.)

2. **Capped Bell–Evans–Polanyi Law (`'bep_cap'`, Used in `reference_bep` / `M0-BEP` and `+BEP` Structures):** (`bep_cap`):

$$\Delta G^{\ddagger}_{f,j}(x_j, g_j) = \max(g_j,\ g_j + \alpha\, x_j) \tag{31}$$

   - In `M0-BEP` (`reference_bep`): $\alpha = 0.5$ and $g = 1.15$ eV for all 9 reactions.
   - In `+BEP` structures (`with_bep_rule`): $\alpha = 1.0$, so every downhill step ($\Delta G_{\text{rxn},j} \le 0$) has barrier $\Delta G^{\ddagger}_{f,j} = g_j$, and every uphill step ($\Delta G_{\text{rxn},j} > 0$) has barrier $\Delta G^{\ddagger}_{f,j} = g_j + \Delta G_{\text{rxn},j}$, restoring reversal invariance.

Step 5.2 · Forward Rate Constant $k_{f,j}(T)$ via Eyring Transition-State Theory (`kinetics/microkinetics/rates.py`)

In `calculate_rate_constants`, the forward rate constant at temperature T is calculated from $\Delta G^{\ddagger}_{f,j}(T)$ via the Eyring equation (with transmission coefficient $\kappa = 1$):

$$k^{\mathrm{TST}}_{f,j}(T) = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^{\ddagger}_{f,j}(T)}{RT}\right) \tag{32}$$

where $\frac{k_B T}{h} \approx 6.212 \times 10^{12}$ s^{-1} at 298.15 K (and concentration standard-state powers $(c^*)^{1-m_j}$ equal 1 M$^{1-m_j}$ in molar units).

If a solvent viscosity η (`viscosity_Pa_s`) is provided, bimolecular reactions ($\sum_i \nu^{\mathrm{react}}_{i,j} = 2$) are placed in series with the Smoluchowski diffusion limit $k_D(T) = \frac{8RT}{3\eta}$ (in M^{-1}s^{-1}) via the **Collins–Kimball rule**:

$$k_{f,j}(T) = \left(\frac{1}{k^{\mathrm{TST}}_{f,j}(T)} + \frac{1}{k_D(T)}\right)^{-1} \tag{33}$$

Step 5.3 · Reverse Rate Constant $k_{r,j}(T)$ via Thermodynamic Detailed Balance

The reverse rate constant $k_{r,j}(T)$ is **never** parameterized independently. It is strictly fixed by thermodynamic detailed balance using $K_{\mathrm{eq},j}(T) = \exp(-\Delta G_{\mathrm{rxn},j}(T)/RT)$:

$$k_{r,j}(T) = \frac{k_{f,j}(T)}{K_{\mathrm{eq},j}(T)} = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^{\ddagger}_{f,j}(T) - \Delta G_{\mathrm{rxn},j}(T)}{RT}\right) = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^{\ddagger}_{r,j}(T)}{RT}\right) \tag{34}$$

Because $k_{r,j}(T) = k_{f,j}(T)/K_{\mathrm{eq},j}(T)$ holds for every reaction j, the net rate $r_j(\mathbf{C}, T)$ vanishes if and only if $\prod_i C_i^{s_{i,j}} = K_{\mathrm{eq},j}(T)$. Consequently, **any isothermal trajectory $\mathbf{C}(t)$ integrated as $t \to \infty$ is mathematically guaranteed to converge to the exact same equilibrium composition $\mathbf{C}^{\mathbf{eq}}(T)$ computed by the thermodynamic solver**.

Step 5.4 · Elementary Net Reaction Rates and Coupled ODE System (`kinetics/reactor/engine.py`)

In `MassActionSystem`, concentrations are clamped to non-negative values $\tilde{C}_i = \max(C_i, 0)$ (with buffered species such as neat `EC` held at their initial concentration $\tilde{C}_{\mathrm{EC}} = C_{\mathrm{EC},0}$ when `buffered_species=('EC',)`).

1. **Net rate of elementary step j (`calculate_net_rates`):**

$$r_j(\mathbf{C}, T) = k_{f,j}(T) \prod_{i=1}^{N_{\mathrm{sp}}} \tilde{C}_i^{\,\nu^{\mathrm{react}}_{i,j}} - k_{r,j}(T) \prod_{i=1}^{N_{\mathrm{sp}}} \tilde{C}_i^{\,\nu^{\mathrm{prod}}_{i,j}} \qquad [\mathrm{mol\ L^{-1}\ s^{-1}}] \tag{35}$$

   For instance, for R1 (TMSPA + $H_2O \rightleftharpoons$ BMSPA + TMSOH) and R4 (2 TMSOH $\rightleftharpoons$ HMDSO + H_2O):

$$r_1(t) = k_{f,1}(T)\, C_{\mathrm{TMSPA}}(t)\, C_{\mathrm{H_2O}}(t) - k_{r,1}(T)\, C_{\mathrm{BMSPA}}(t)\, C_{\mathrm{TMSOH}}(t) \tag{36}$$

$$r_4(t) = k_{f,4}(T)\, \left(C_{\mathrm{TMSOH}}(t)\right)^2 - k_{r,4}(T)\, C_{\mathrm{HMDSO}}(t)\, C_{\mathrm{H_2O}}(t) \tag{37}$$

2. **Coupled Species ODE System (`rhs`):**

$$\frac{d\mathbf{C}}{dt} = S\, \mathbf{r}(\mathbf{C}, T), \qquad \mathbf{C}(0) = \mathbf{c}_0 \tag{38}$$

   (For any buffered species b, row b of S is zeroed out so $\frac{dC_b}{dt} = 0$.)

3. **Exact Analytical Jacobian (`jacobian`):** Because elementary rate constants $k_{f,j}(T)$ across different reaction families can span 10–15 orders of magnitude, the ODE system is stiff. To avoid finite-difference overflow during implicit integration (`BDF` or `Radau`), `MassActionSystem` supplies the exact analytical Jacobian $J(\mathbf{C}) = \frac{\partial(S\mathbf{r})}{\partial \mathbf{C}} = S\frac{\partial \mathbf{r}}{\partial \mathbf{C}}$:

$$\frac{\partial r_j}{\partial C_i} = k_{f,j}\,\nu_{i,j}^{\text{react}}\,\tilde{C}_i^{\,\nu_{i,j}^{\text{react}}-1}\prod_{m\neq i}\tilde{C}_m^{\,\nu_{m,j}^{\text{react}}} - k_{r,j}\,\nu_{i,j}^{\text{prod}}\,\tilde{C}_i^{\,\nu_{i,j}^{\text{prod}}-1}\prod_{m\neq i}\tilde{C}_m^{\,\nu_{m,j}^{\text{prod}}} \tag{39}$$

Step 5.5 · Piecewise-Isothermal Tube History Integration (`kinetics/reactor/protocol.py`)

Each experimental NMR tube s undergoes a sequence of isothermal segments $((T_1, \Delta t_1), (T_2, \Delta t_2), \ldots, (T_M, \Delta t_M))$ starting from mixing ($t = 0$):

- An initial room-temperature pre-history of duration t_{age} (the time between sample mixing and the first NMR acquisition, e.g., scenario readings `'10 min'`, `'1 h'`, or `'8 h'` when unrecorded),
- Alternating room-temperature storage intervals and elevated-temperature heating holds (e.g., 50 °C or 80 °C water/oil bath episodes).

`simulate_history` integrates segment-by-segment using `scipy.integrate.solve_ivp`:

1. At the start of segment m $[t_{m-1}, t_m]$ with $t_m = t_{m-1} + \Delta t_m$, rate constants $(\mathbf{k}_f(T_m), \mathbf{k}_r(T_m))$ are evaluated (and cached by temperature T_m).
2. The ODE system is integrated from the terminal state of the previous segment $\mathbf{C}(t_{m-1})$, evaluating $\mathbf{C}(t_{s,k})$ at every NMR spectrum acquisition timestamp $t_{s,k} \in (t_{m-1}, t_m]$.

Step 5.6 · Summary Comparison: Thermodynamics vs. Kinetics

Aspect	**Thermodynamics (`kinetics/thermo/equilibrium.py`)**	**Microkinetics (`kinetics/microkinetics/` + `reactor/`)**
Physical question	What is the final equilibrium composition $\mathbf{C}^{\text{eq}}(T)$ as $t \to \infty$?	What is the instantaneous composition $\mathbf{C}(t)$ at every second t along $T(t)$?
Energy parameters	Species free energies $G_{\text{sol},i}(T) \implies \Delta G_{\text{rxn},j}(T)$ only	$\Delta G_{\text{rxn},j}(T)$ **plus** intrinsic barriers $g_{\text{fam}} = E_{0,\text{fam}} \implies \Delta G_{f,j}^{\ddagger}(T)$
Mathematical problem	Algebraic system: mass-action laws + moiety conservation (solved via 3D/5D convex dual $\boldsymbol{\lambda}$)	Dynamical system: coupled stiff ODEs $\frac{d\mathbf{C}}{dt} = S\,\mathbf{r}(\mathbf{C}, T(t))$ from $\mathbf{C}(0) = \mathbf{c}_0$
Dependence on sample age t_{age}	None (independent of time and heating history)	Directly depends on t_{age} and the timing/duration of every heating episode
Role of redundant cycles (`R5`–`R7`, `N10`–`N15`)	Algebraically redundant ($K_5 = K_1K_4$ adds no new constraint on $\mathbf{C}^{\text{eq}}$)	Active parallel kinetic pathways that accelerate silyl transfer and hydrolysis fluxes

6 Mapping Concentrations to NMR Observables & Error Models (`kinetics/fitting/residuals.py`)

6.1 NMR Window Definitions (`SHARE_SPECIES`)

An NMR spectrum of nucleus $\mathcal{N} \in \{^{31}\text{P}, ^{13}\text{C}, ^{29}\text{Si}, ^{1}\text{H}\}$ does not measure molar concentrations directly; it measures the integrated peak area within chemical-shift windows $w \in \mathcal{W}_\mathcal{N}$. Each species i contributes $n_{\mathcal{N},w,i}$ magnetically equivalent nuclei (or silyl groups) per molecule to window w as defined in `SHARE_SPECIES`:

Nucleus $\mathcal{N}$	**Window Name w**	**Contributing Species & Multiplicity $n_{\mathcal{N},w,i}$**	**Physical Meaning of Window**
^{31}**P**	`TMSPA`	`TMSPA`: 1	Intact tris(trimethylsilyl) phosphate
	`BMSPA`	`BMSPA`: 1	Mono-hydrolyzed phosphate ester
	`MMSPA`	`MMSPA`: 1	Di-hydrolyzed phosphate ester
	`H3PO4`	`H3PO4`: 1	Fully hydrolyzed phosphoric acid
^{13}**C** *(and ^{29}Si)*	`TMSOH`	`TMSOH`: 1	Trimethylsilanol $-\text{Si}(\text{CH}_3)_3$
	`HMDSO`	`HMDSO`: 2	Hexamethyldisiloxane (2 $-\text{Si}(\text{CH}_3)_3$ groups per molecule)
	`TMSOEG`	`TMSOEG`: 1, `TMSOdiEG`: 1	Ring-opened ethylene glycol silyl ethers
	`P-silyl`	`TMSPA`: 3, `BMSPA`: 2, `MMSPA`: 1	Total silyl groups still bound to phosphorus
^{1}**H**	`P-bound silyl`	`TMSPA`: 3, `BMSPA`: 2, `MMSPA`: 1	Silyl $-\text{Si}(\text{CH}_3)_3$ protons on phosphate esters
	`free silyl`	`TMSOH`: 1, `HMDSO`: 2, `TMSOEG`: 1, `TMSOdiEG`: 1	Silyl $-\text{Si}(\text{CH}_3)_3$ protons cleaved from phosphate

6.2 Predicted Window Shares vs. Millimolar Concentrations

Given a concentration vector $\mathbf{C}$ (either $\mathbf{C}^{\text{eq}}(T)$ from the thermodynamic solver or $\mathbf{C}(t_k)$ from the kinetic reactor at spectrum time t_k), the nuclear abundance in window $w \in \mathcal{W}_\mathcal{N}$ is:

$$A_{\mathcal{N},w}(\mathbf{C}) = \sum_{i \in w} n_{\mathcal{N},w,i}\, \max(C_i, 0) \tag{40}$$

Two observable conventions are supported:

1. **Normalized Window Shares (`error_model='kinetic'`, `_shares_from_state`):** Divides each window's nuclear abundance by the sum over all active windows $\mathcal{W}_{\text{block}}$ measured in that spectrum:

$$\hat{y}_w(\mathbf{C}) = \frac{A_{\mathcal{N},w}(\mathbf{C})}{\sum_{w' \in \mathcal{W}_{\text{block}}} A_{\mathcal{N},w'}(\mathbf{C})} \tag{41}$$

   - For ^{31}**P**, $\sum_{w'} A_{^{31}\text{P},w'} = C_{\text{TMSPA}} + C_{\text{BMSPA}} + C_{\text{MMSPA}} + C_{\text{H}_3\text{PO}_4} = C_{\text{TMSPA},0}$, so $\hat{y}_w = C_w / C_{\text{TMSPA},0}$ is the exact mole fraction of phosphorus in ester w.
   - For ^{13}**C** / ^{29}**Si** / ^{1}**H**, $\sum_{w'} A_{\mathcal{N},w'} = 3\, C_{\text{TMSPA},0}$ (total silyl groups), so $\hat{y}_w$ is the fraction of all $-\text{SiMe}_3$ groups residing in window w.

2. **Millimolar Concentrations (`error_model='nmr'` in `fit_thermo_rev01`, `_concentration_block`):** Converts the experimental shares $y_{\mathrm{NMR},w}$ and model predictions into millimolar (mM) concentrations using the known total nuclear concentration at mixing $A_{\mathcal{N},\mathrm{tot}}(\mathbf{c}_0) = \sum_{w\in\mathcal{W}_{\mathrm{block}}}\sum_{i\in w} n_{\mathcal{N},w,i}c_{0,i}$ and window divisor d_w ($d_w = n_{\mathcal{N},w,i}$ for a single-species window such as $d_{\mathrm{HMDSO}} = 2$, or $d_w = 1$ for a multi-species group window):

$$\hat{c}_w(\mathbf{C}) = 1000 \cdot \frac{A_{\mathcal{N},w}(\mathbf{C})}{d_w} \quad [\mathrm{mM}], \qquad c_{\mathrm{NMR},w} = 1000 \cdot y_{\mathrm{NMR},w} \cdot \frac{A_{\mathcal{N},\mathrm{tot}}(\mathbf{c}_0)}{d_w} \quad [\mathrm{mM}] \tag{42}$$

6.3 Measurement Error Model & Residual Whitening (`_lab_block`, `_whiten`)

Each integrated window w in spectrum k carries two raw spectral uncertainty estimates from `scripts/build_observables.py`:

- $\sigma_{\mathrm{noise},k,w}$: thermal spectral noise propagated through the integration window width,
- $\sigma_{\mathrm{baseline},k,w}$: variation of the window integral under alternative baseline corrections.

In addition, the kinetic error model defines a regularizing error floor (`DEFAULT_ERROR_FLOOR`):

$$\sigma_{\mathrm{floor}}(y) = s_{\mathrm{floor}} \cdot \max(\sigma_{\mathrm{abs}},\ \sigma_{\mathrm{rel}} \cdot \max(y, 0)) \tag{43}$$

1. **Single / Non-Replicate Spectra:** All error components are independent across spectra:

$$\sigma_{\mathrm{ind},k,w} = \sqrt{\sigma^2_{\mathrm{noise},k,w} + \sigma^2_{\mathrm{baseline},k,w} + \sigma_{\mathrm{floor}}(y_{k,w})^2} \tag{44}$$

   (In `error_model='nmr'`, $\sigma_{floor} = 0$, so $\sigma_{tot,k,w} = \sqrt{\sigma^2_{noise,k,w} + \sigma^2_{baseline,k,w}}$ scaled to mM.)

2. **Replicate Room-Temperature Spectra ($N_{\mathbf{rep}} \geq 3$ at $T < 30\ ^\circ\mathbf{C}$):** Repeated room-temperature spectra of the same tube represent 1 measurement of the mean composition plus $N_{\mathrm{rep}} - 1$ measurements of drift. The independent noise is scaled by the empirical Birge ratio $B = \max(1, B_{\mathrm{rep}})$, while the baseline uncertainty and floor form a shared covariance component $\sigma^2_{\mathrm{common},w}$:

$$\Sigma_w = \mathrm{diag}(\sigma^2_{\mathrm{ind},1,w}, \ldots, \sigma^2_{\mathrm{ind},N_{\mathrm{rep}},w}) + \sigma^2_{\mathrm{common},w}\,\mathbf{1}\mathbf{1}^T = L_w L_w^T \tag{45}$$

   where L_w is the lower-triangular Cholesky factor used to whiten the residual vector via $L_w^{-1}(\hat{\mathbf{y}}_w - \mathbf{y}_w)$.

3. **Simplex Sum-Constraint Scaling:** Because the $K = |\mathcal{W}_{\mathrm{block}}|$ window shares of a spectrum sum to 1 ($\sum_{w=1}^{K} y_{k,w} = 1$), there are only $K - 1$ independent degrees of freedom per spectrum. In `_whiten`, the K standardized residuals are multiplied by $\sqrt{\frac{K-1}{K}}$ so their sum of squares equals the exact χ^2 of $K - 1$ independent variables.

7 Parameter Estimation & Two-Stage Fitting Workflow (`kinetics/fitting/`)

7.1 Stage 1: Thermodynamic Fit at Equilibrium (`kinetics/fitting/thermo_fit.py`)

When an NMR tube has finished reacting at room temperature, its spectrum stops changing over time. Because an equilibrium state depends **only on the reaction free energies** $\Delta G_{\mathbf{rxn},j}$ (and is completely independent of activation barriers E_0 and unknown mixing-to-acquisition ages t_{age}), `thermo_fit.py` calibrates the solvation energies first using the algebraic equilibrium solver:

1. **Automatic Equilibrium Plateau Detection (`find_equilibrium_points`):** A room-temperature, quantitative spectrum (^{31}P with NOE off and $d_1 \geq 5$ s; ^{29}Si non-INEPT) with ≥ 2 windows is classified as being at equilibrium if it belongs to a trailing plateau of at least `min_spectra = 2` spectra spanning at least `min_span_h = 24 h` whose window shares agree within `max_abs_z = 3.0` of the noise-weighted mean of the others.

2. **Parameters & Box Constraints (`ThermoProblem`):** The free parameters are the solvation energy shifts $\delta_i \equiv \delta E_{\text{solv},i}$ for the 7 species of `R1`–`R4`:

$$\text{free_species} = \{\text{TMSPA, BMSPA, MMSPA, } H_3PO_4\text{, } H_2O\text{, TMSOH, HMDSO}\} \tag{46}$$

   bounded within $\pm\xi$ MD standard errors ($\xi \in \{1, 2, 3\}$):

$$-\xi\,\sigma_i \leq \delta E_{\text{solv},i} \leq +\xi\,\sigma_i \tag{47}$$

   where $\sigma_i = \sigma_{\text{solute},i}$ (`sigma_solv='solute'`) or $\sigma_i = \sqrt{\sigma^2_{\text{solute},i} + \sigma^2_{\text{solvent}}}$ (`sigma_solv='total'`), plus any unrecorded initial water concentration (`c0 H2O <tube>`).

3. **Identifiability of $\Delta G_{\mathbf{rxn},j}$ vs. Individual $\delta E_{\mathbf{solv},i}$:** Because equilibrium concentrations depend only on $\Delta G_{\text{rxn},j} = \Delta G^{\text{comp}}_{\text{rxn},j} + \sum_i s_{i,j}\delta E_{\text{solv},i}$ for the 4 independent reactions `R1`–`R4`, the data constrain at most **4 linear combinations** of the 7 species shifts $\delta E_{\text{solv},i}$ (the remaining $7-4=3$ directions lie in the nullspace $A = \text{null}(S^T)$ and do not change any concentration). Therefore, `tabulate_reaction_energies` and `calculate_energy_ranges` report the fitted reaction energies $\Delta G_{\text{rxn},j}$ and equilibrium constants $K_{\text{eq},j}$ along with their profile confidence envelopes.

7.2 Stage 2: Microkinetic Fit Across Candidate Structures (`kinetics/fitting/candidates.py` & `estimation.py`)

With the thermodynamics established, `FitStructure` defines a nested hierarchy of kinetic models tested against the time-resolved NMR spectra:

Structure Name	Barrier Law	Barrier Parameters Fitted	Reaction Energies $\Delta G_{\mathbf{rxn}}$
`M0-BEP`	Capped BEP ($\alpha = 0.5$)	1 shared barrier (`g all reactions`)	Fixed at computed values
`M0-Marcus`	Marcus ($\lambda = 4g$)	1 shared barrier (`g all reactions`)	Fixed at computed values
`M1`	Marcus ($\lambda = 4g$)	4 family barriers (`g hydrolysis`, `g transfer`, `g condensation`, `g solvent_attack`)	Fixed (or handed over from `thermo_fit`)
`M1-split`	Marcus ($\lambda = 4g$)	5 barriers (`R1` split from `R2, R3`: `g hydrolysis_R1`, `g hydrolysis_R23`, `g transfer`, `g condensation`, `g solvent_attack`)	Fixed
`M3-split`	Marcus ($\lambda = 4g$)	Same 5 barriers as `M1-split`	ΔG_{rxn} of `R1`–`R4` freed with Gaussian covariance prior $\mathbf{d}^T\Sigma^{-1}\mathbf{d}$
`<Name>+BEP`	Capped BEP ($\alpha = 1.0$)	Same barrier grouping as `<Name>`	Same as `<Name>`

(When ΔG_{rxn} of `R1`–`R4` are freed by shifts $\mathbf{d} = (\Delta G_1 - \Delta G_1^{comp}, \ldots, \Delta G_4 - \Delta G_4^{comp})^T$ in `M3-split`, `calculate_species_shifts` maps $\mathbf{d}$ onto minimum-norm species shifts $\boldsymbol{\delta} = S_{sp}A^T(AS_{sp}A^T)^{-1}\mathbf{d}$ so `R5`–`R7` automatically follow by Wegscheider cycle closure and `R8`–`R9` retain their computed ΔG_{rxn}.)

7.3 Global-to-Local Optimization and Profile Likelihoods (`estimation.py`)

1. **Box Coordinate Transformation:** Every free parameter $\theta_p \in [\theta_p^{\min}, \theta_p^{\max}]$ is mapped to unit box coordinates $u_p = 1 + \frac{\theta_p - \theta_p^{\min}}{\theta_p^{\max} - \theta_p^{\min}} \in [1, 2]$ so finite-difference Jacobian steps (`DIFF_STEP = 1e-3`) correspond to the same relative fraction (0.1%) of every parameter's search interval.

2. **Two-Phase Optimization (`fit_structure`):**

   - **Global Sobol Screen:** Evaluates $\chi^2(\mathbf{u})$ across N_{screen} scrambled Sobol quasi-random points in $[1, 2]^n$ (with loose ODE tolerance `rtol = 1e-6`, `atol = 1e-13`) and selects the N_{starts} lowest-χ^2 points separated by at least $0.05\sqrt{n}$ in box distance.
   - **Local Trust-Region Reflective (`trf`) Polish:** Launches bounded nonlinear least squares (`scipy.optimize.least_squares`) on the whitened residual vector $\mathbf{r}(\mathbf{u})$ from each start point, followed by a final high-precision polish using `REPORT_SOLVER` (`rtol = 1e-8`, `atol = 1e-14`).

3. **Acceptance Criterion & Confidence Intervals (`calculate_profile`, `find_confidence_interval`):**

   - A candidate model **fits** a sample set if every standardized residual satisfies $|z_k| \leq 3.0$ (`FIT_RULE_MAX_Z = 3.0`).
   - 1D profile likelihoods fix one parameter θ_p along a grid around $\hat{\theta}_p$ and re-minimize χ^2 over all remaining parameters; the 95% profile confidence interval is bounded by the crossings where $\Delta\chi^2(\theta_p) = \chi^2_{\text{profile}}(\theta_p) - \chi^2_{\min} = 3.84$ (`DELTA_CHI2_95`).